

From Obstruction to Opportunity: Revisiting the Timing of Esophageal Stents in Definitive Cancer Management: Case Report

Dr Bessy Thomas
University of Witwatersrand
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Declaration of Interest

- Nil to declare
- Employment: Consultant Radiation Oncologist at Charlotte Maxeke Johannesburg Academic Hospital.

Case Summary

- 59-year-old male, no known comorbidities, with Oesophageal Cancer cT3N1M0 (SCC) now for neo-adjuvant chemoradiation followed by surgery.

History

- 59-year-old male referred with a 1 year history of progressive dysphagia and odynophagia requiring immediate stent placement on admission. Currently tolerating a soft diet. Associated loss of weight (>10%) and appetite, no hoarseness of voice or cough.
- Past Medical History: HIV negative, no comorbidities, no history of caustic ingestion.
- Social history: Strong positive smoking history 40 pack-year, no family history of cancers.

CLINICAL EXAMINATION

- Alert and well, Vitals normal, BMI: 16,7kg/m, BSA: 1.73m²
- ECOG 0
- General exam: no palpable lymphadenopathy, not pale, no jaundice or pedal oedema
- Systems exam: unremarkable

IMAGING

- Biochemical: FBC/U&E/CMP and LFT normal (Hb 12,7g/dl, Urea 2.9mmol/L, Creatinine 66 umol/L, albumin 43g/dl)
- GI scope: lesion noted at 40cm from the incisors, unable to pass scope further. Guidewire passed through stomach under screening and 24mm x 80mm stent placed.
- *Endoscopic ultrasound not performed due to equipment not available*
- Biopsy: invasive moderately differentiated, keratinizing squamous cell carcinoma with high grade dysplasia. No lymphovascular or perineural invasion noted. Malignancy involves underlying muscularis.

IMAGING

- CT CAP: Distal gastroesophageal circumferential mural thickening. Mucosal/submucosal enhancement with impression of adventitia involvement. No luminal occlusion(stent migration into stomach), Poor planes of separation from adjacent structures at T11 vertebra. Enhancing mediastinal lymph nodes- AP window (SAD 9.3x LAD 11mm) and carina (SAD 7.3 x LAD 9.4mm). Thickened and enhancement of gastric cardia. No direct invasion of the diaphragm. No distant metastasis.
- Stage T3N1M0
 - PET/CT: Not done. Resource constraints.



MANAGEMENT

- Patient assessed at MDT meeting with upper GI surgeon, radiation oncologist, medical oncologist and axillary services (oncology nurse, dietician and social worker)
- *Cancer of the lower thoracic oesophagus (SCC), at least T3N1M0.*
- *Prognostic factors: bulky tumour, good KPS/ECOG.*
- *Treatment Intent: CURATIVE*
- *For stent removal first before neoadjuvant CCRT followed by definitive surgery.*

- Patient was seen prior to Planning CT for removal of stent. However the tumour had overgrown and had migrated into the stomach and small intestine. It was noted to be too difficult to remove.
- The decision was then to continue with chemo-radiation so as to not further delay patient treatment.

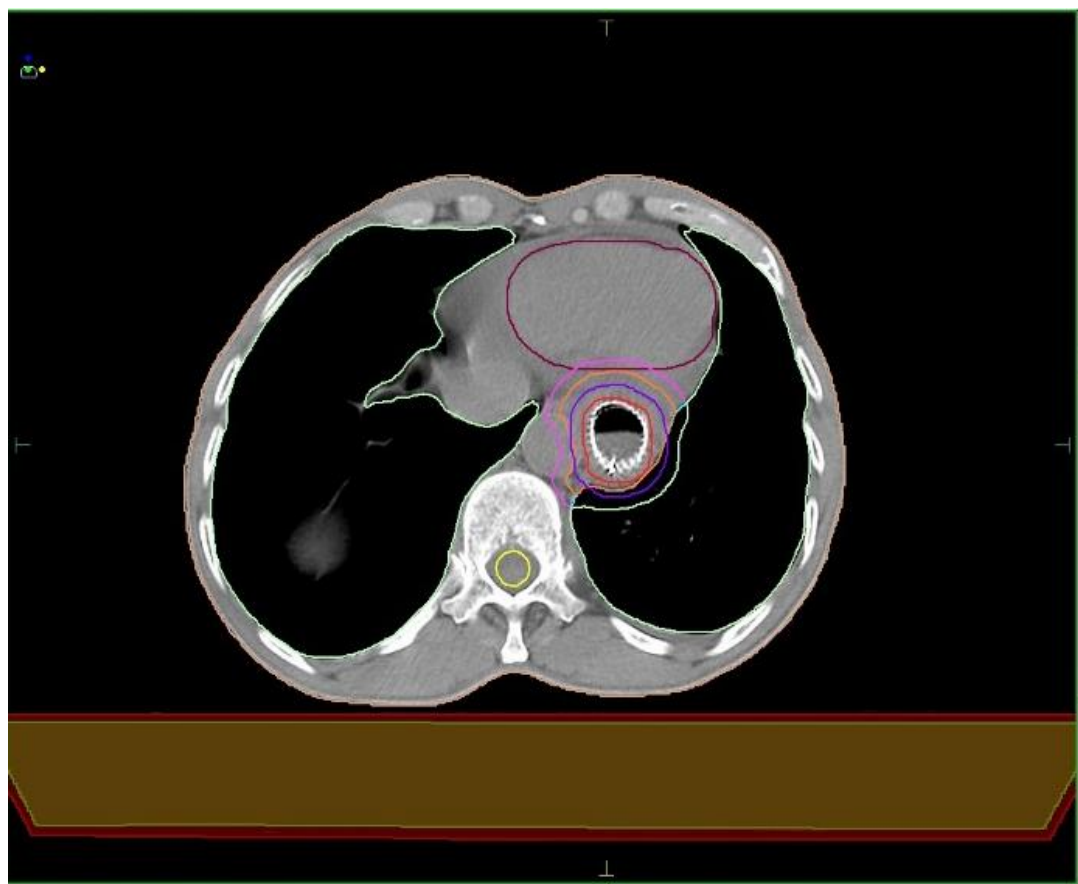
Planning CT

- **Sedation:** Nil
- **Positioning:** Supine. Head first
- **Immobilization** Wing board used with arms above the head
- **CT instructions** Planning CT from angle of mandible to L5, 5mm slices, IV and oral contrast.
- *Motion management techniques not used: 4DCT, ABC DIBH and abdominal compression not available.*

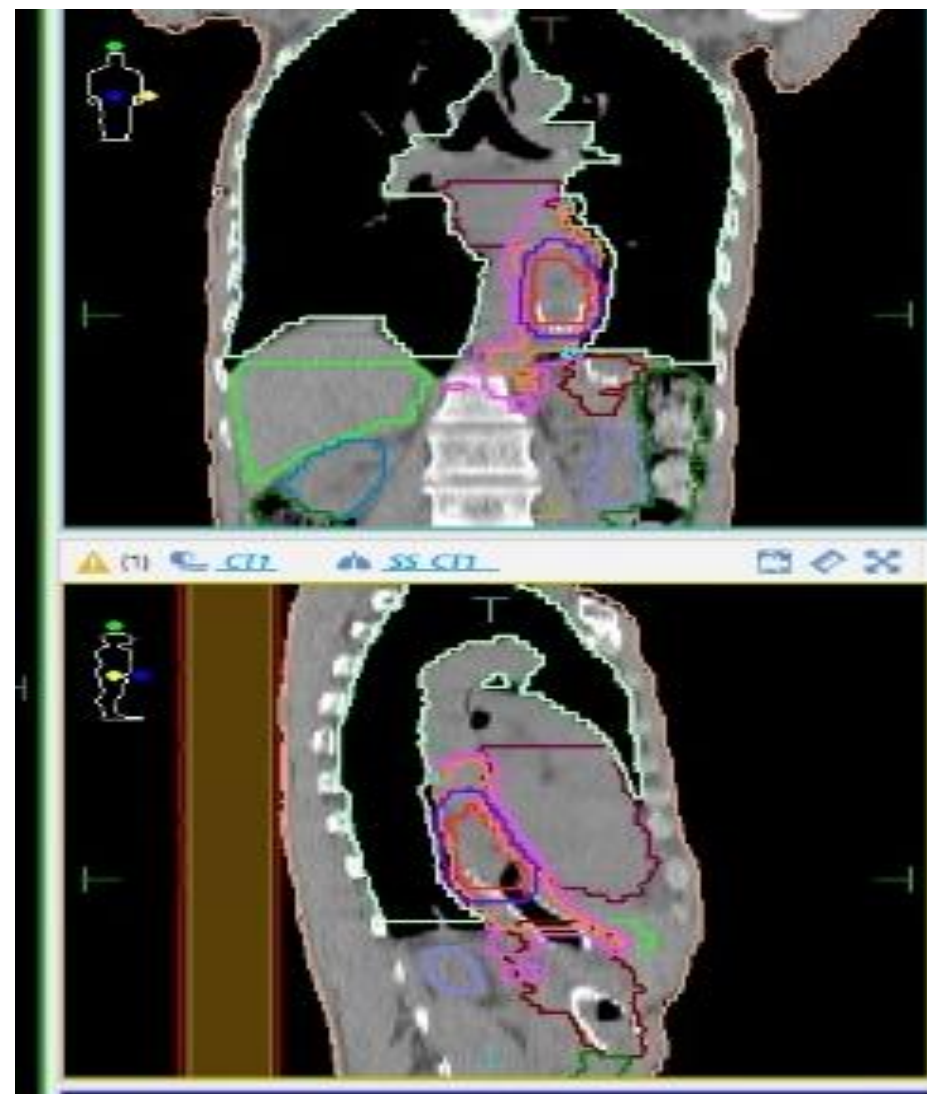
- GTV All visible tumour noted on planning CT. GTV primary and GTV nodes contoured. Diagnostic CT fused and used to aid delineation
- CTV Margin of 3cm superior and inferior used with 1cm circumferentially. Elective nodal coverage included coeliac , periesophageal, mediastinal nodes, lesser curvature and porta hepatis. Margin cropped off vertebrae, lung, vessel and heart.
- PTV:
 - PTV45: 0,5cm margin added to CTV
 - PTV50,4: 0,5cm margin added to GTV
- Organs at risk contoured: bilateral lungs, heart, spinal cord, bilateral kidneys, bowelbag and stomach

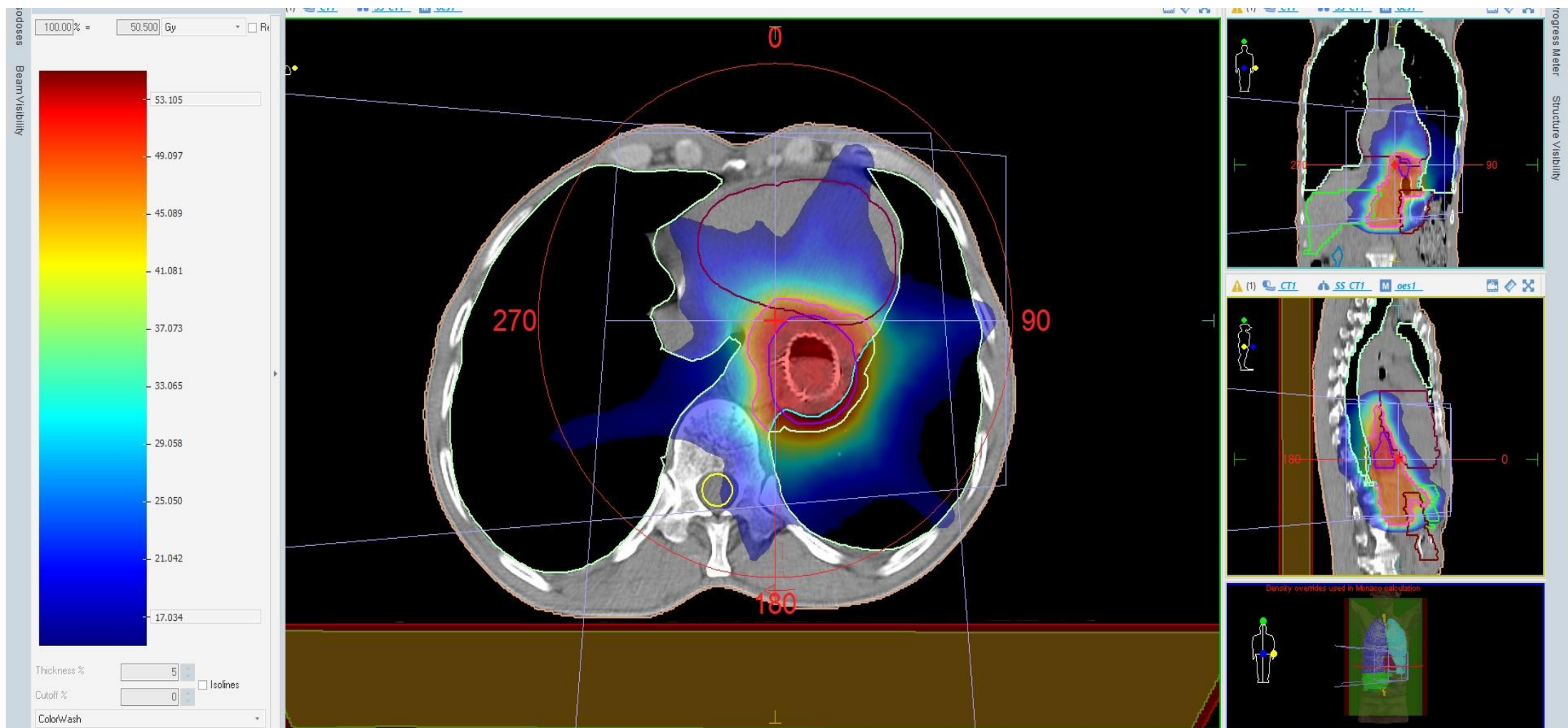
TREATMENT DETAILS

- CCRT:
- Total Dose: 45Gy in 1.8Gy per fraction for a total of 25 fractions to PTV (tumour and elective nodes)
- BOOST: Total Dose: 50,4Gy in 2,016Gy per fraction for a total of 25 fractions to tumour only.
- Concurrent:
 - Cisplatin 80mg/m² IVI 3weekly
 - Capecitabine 825mg/m² po bd for 25 days (days of RT)



**GTV: RED CTV: ORANGE PTV45: PINK PTV50,5:
BLUE**





RT Plan review

D 95 of PTV 50.5	100						
D95 of PTV 45	97.8						
Dmax/Hotspot	105%						
OARs							
Bilateral lung		Heart		Kidneys		Liver	
V40 < 10%	0.52	V30 < 30%	11.9	V20 < 33%	RT: 0	V30 < 33%	6.9
V30 < 15%	2.17	MD < 26 Gy – 30Gy	19.16	MD < 16-18 Gy	Rt: 1.6 Gy	MD < 25Gy	13.96Gy
V10 < 40%	28.7			V20 <33%	Lt: 5.05 Gy		
MD < 20 Gy	7 Gy			MD < 16-18Gy	Lt: 5.9Gy		
Bowel Bag		Spinal cord		Stomach			
Dmax < 54Gy	31	Dmax < 45Gy	22.7	Dmax <54Gy	48.6		
V45 < 195cc	0			MD < 45Gy	20.8		

- The stent had migrated into the stomach, unable to be removed prior to treatment.
- The patient received radiation from 13/11/2024 to 19/12/2024 with no breaks in treatment.
- Weekly follow up during treatment: c/o chest pain, dysphagia to soft diet, only tolerating liquid diet
- At 16th fraction, urea noted to be 14.4mmol/L and creatinine 162umol/L. The patient complained of persistent vomiting post chemotherapy. Admitted for IV fluids with improvement in U&Es.

- He had a two-stage oesophagostomy in February 2025.
- Histopath Report: Tumour Size: 4,4 x3,2mm. Tumour midpoint lies in the distal oesophagus, and the tumour involves the distal esophagogastric junction. Margins uninvolved.
- LND: 1/7 nodes. No LVSI. No PNI. ypT2N1M0
- Follow-up: 06/05/2025: Patient stable. Eating well. Gaining weight.

TIMELINE

- Initial symptoms: July 2023
- GI scope, biopsy and stent placement: 4 April 2024
- Staging CT: 24 June 2024
- MDT discussion: 19 August 2024
- Initial consult Radiation Oncology: 26 August 2024
- Treatment: 13 November 2024 – 19 December 2024
- Surgery: February 2025

DISCUSSION POINTS

- Role of Stent placement before definitive treatment
- Ideal candidates for stent placement
- Can dilation be used instead of stenting.
- Alternatives to stent placement in the management of dysphagia
- Role of PEG insertion
- Is stent migration common?

THANK YOU

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